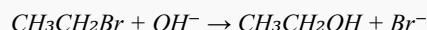


Organic Chemistry Reaction Mechanisms

Problem 1

Provide a complete mechanism for the following reaction, showing all intermediates and electron flow:



- Is this an SN1 or SN2 reaction? Justify your answer.
- What would happen to the rate if the concentration of OH⁻ is doubled?
- What would happen if the leaving group is changed from Br⁻ to F⁻?

Solution:

Mechanism (SN2):

- Nucleophile (OH⁻) attacks the electrophilic carbon from the backside, simultaneously breaking the C-Br bond
- Transition state with pentacoordinate carbon (partial bonds to C, Br, and OH)
- Br⁻ leaves, forming the product alcohol

a) This is an **SN2 reaction** because:

- The substrate is a primary alkyl halide (CH₃CH₂Br)
- The reaction is bimolecular (rate depends on both [RX] and [Nu⁻])
- The mechanism involves a single step with inversion of configuration
- Strong nucleophile (OH⁻) and good leaving group (Br⁻)

b) Since this is SN2, the rate = k[CH₃CH₂Br][OH⁻]. If [OH⁻] is doubled, the rate will also double (linear relationship).

c) If Br⁻ is changed to F⁻, the rate would **decrease significantly** because:

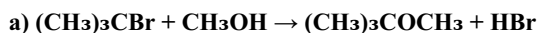
- F⁻ is a poor leaving group (strong C-F bond, high bond energy)
- The rate of SN2 reactions is highly sensitive to leaving group ability
- Br⁻ is a much better leaving group than F⁻ due to weaker C-Br bond and greater stability of Br⁻ ion

Problem 2

Predict the major product(s) for each of the following reactions and provide mechanisms:

- (CH₃)₃CBr + CH₃OH →
- CH₃CH=CH₂ + HBr →
- Cyclohexene + Br₂ →

Solution:



Mechanism (SN1):

1. Tertiary bromide undergoes heterolysis to form tert-butyl carbocation and Br^-
2. CH_3OH (weak nucleophile) attacks the carbocation to form $(\text{CH}_3)_3\text{COCH}_3$
3. Proton transfer from protonated ether to Br^- gives final product

This is SN1 because tertiary alkyl halide favors carbocation formation.



Mechanism (Electrophilic Addition, Markovnikov):

1. HBr protonates alkene to form secondary carbocation ($\text{CH}_3\text{CH}^+\text{CH}_3$)
2. Br^- attacks carbocation to form 2-bromopropane

Markovnikov's rule: H adds to carbon with more hydrogens.



Mechanism (Electrophilic Addition, Anti addition):

1. Br_2 attacks alkene to form bromonium ion intermediate
2. Br^- attacks from opposite side (anti addition) to form trans-dibromide

The product is racemic trans-1,2-dibromocyclohexane.

Problem 3

For the following molecule, identify all chiral centers and determine the maximum number of stereoisomers possible:



Draw all stereoisomers and label each as enantiomer or diastereomer.

Solution:

Structure analysis: $\text{HOOC}-\text{CH}(\text{OH})-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COOH}$

This is $\text{HOOC}-\text{CH}(\text{OH})-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COOH}$

Chiral centers: C_2 ($\text{CH}(\text{OH})$) and C_4 ($\text{CH}(\text{NH}_2)$)

Maximum stereoisomers: 2^n where n = number of chiral centers

Maximum = $2^2 = 4$ stereoisomers

Stereoisomers:

- **(2R, 4R):** Both chiral centers have R configuration
- **(2S, 4S):** Both chiral centers have S configuration (enantiomer of 2R,4R)
- **(2R, 4S):** Opposite configurations (diastereomer)
- **(2S, 4R):** Opposite configurations (diastereomer, enantiomer of 2R,4S)

Relationships:

- (2R,4R) and (2S,4S) are enantiomers
- (2R,4S) and (2S,4R) are enantiomers
- All other pairs are diastereomers

If the molecule has no internal plane of symmetry and no meso form, all 4 stereoisomers are distinct.

Problem 4

Explain why 1,3-butadiene undergoes 1,4-addition with HBr at elevated temperatures, while 1,2-addition is favored at lower temperatures. Provide energy diagrams and discuss the thermodynamic vs. kinetic control.

Solution:

Reaction: 1,3-butadiene + HBr

At low temperature (kinetic control):

- 1,2-addition product (3-bromo-1-butene) forms faster
- This is the kinetically favored product (lower activation energy)
- The transition state leading to 1,2-addition is more stable (more substituted allylic carbocation)

At high temperature (thermodynamic control):

- 1,4-addition product (1-bromo-2-butene) is more stable
- Higher temperature allows equilibrium to be established
- 1,4-product has more stable double bond (disubstituted vs. monosubstituted)
- System reaches thermodynamic equilibrium

Energy Diagram Explanation:

- 1,2-addition: Lower E_a (kinetic product), but higher ΔG (less stable)
- 1,4-addition: Higher E_a (slower formation), but lower ΔG (more stable)
- At low T: Reaction doesn't have enough energy to reverse, so kinetic product accumulates
- At high T: Reaction can reverse, allowing system to reach thermodynamic minimum (1,4-product)

This is a classic example of kinetic vs. thermodynamic control in organic reactions.